Appendix A - Hunter Marine Park Natural Values Report (stereo-BRUV)

Temperate-east Marine Parks Network

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Natural Values

Understanding of natural values

Benthic ecosystem extent

Observations

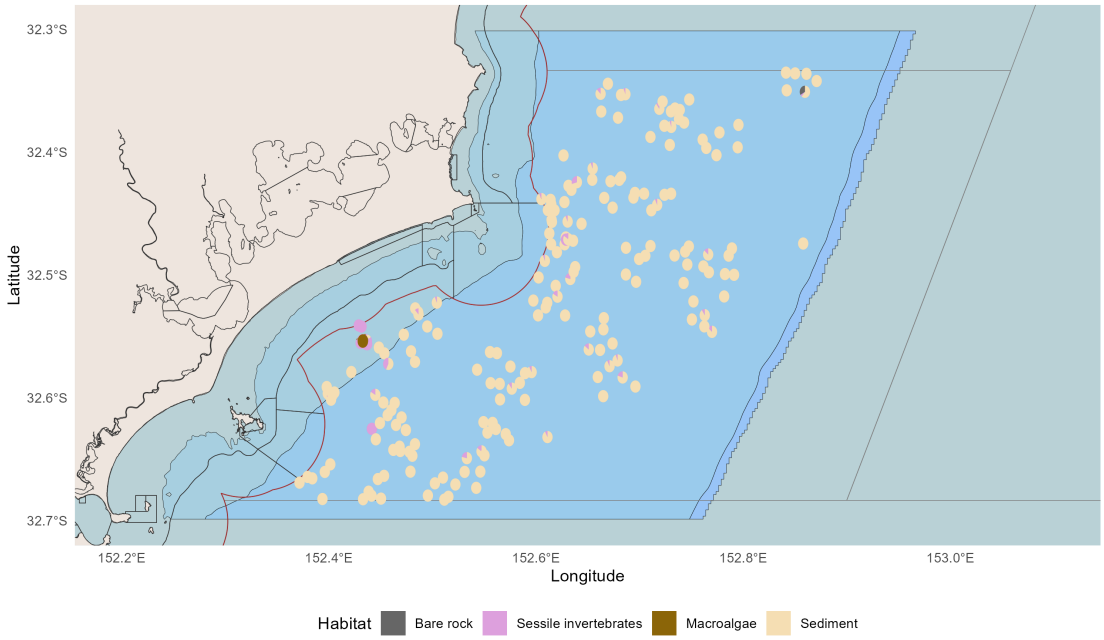


Figure A.1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30, 70 and 200 m are shown.

Dominant habitat types

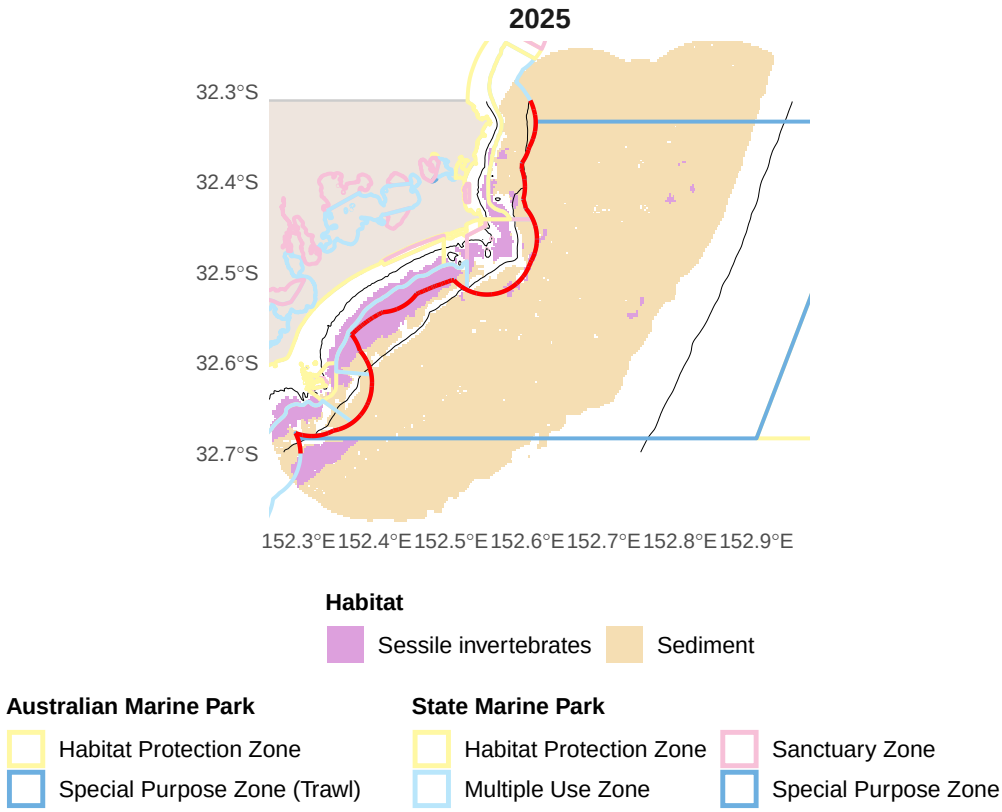


Figure A.2: Predicted dominant habitat type from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30, 70 and 200 m are shown.

Habitat distribution and probability

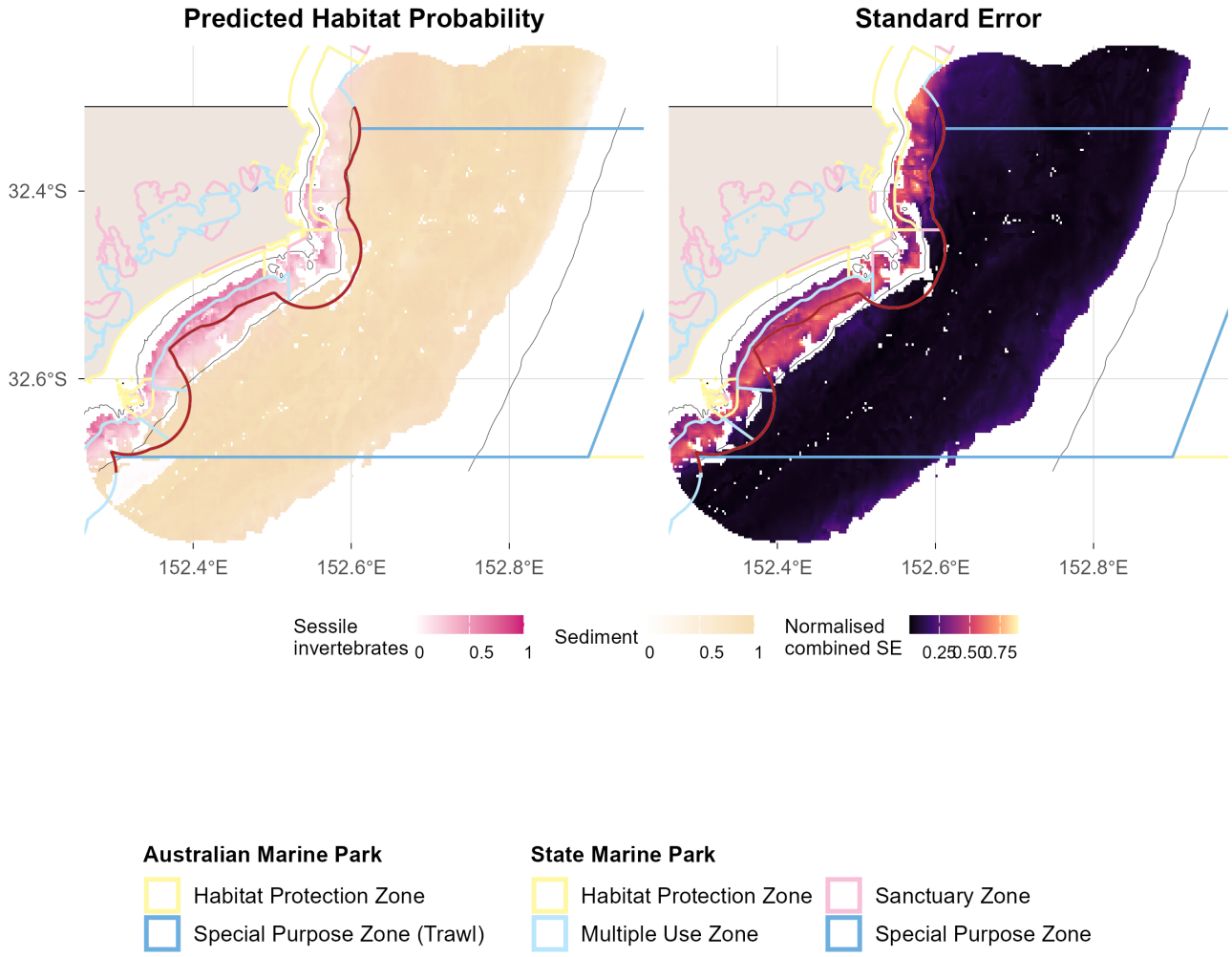


Figure A.3: Predicted habitat distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30, 70 and 200 m are shown.

Benchmarks of benthic natural values

Sediment

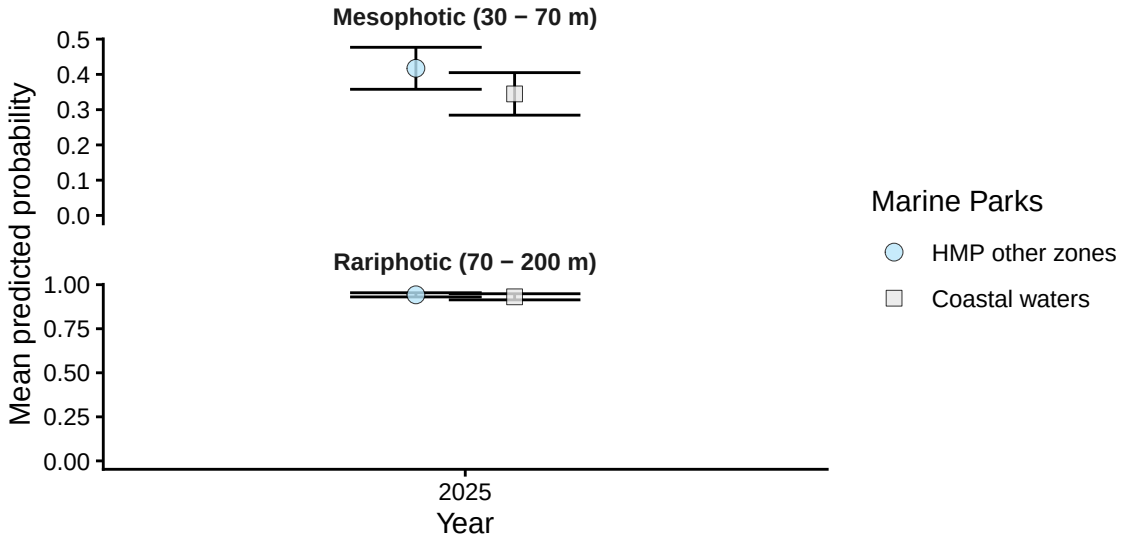


Figure A.4.1: Sediment: temporal plots by depth contour from stereo-BRUV groundtruthing. HMP = Hunter Marine Park (Commonwealth).

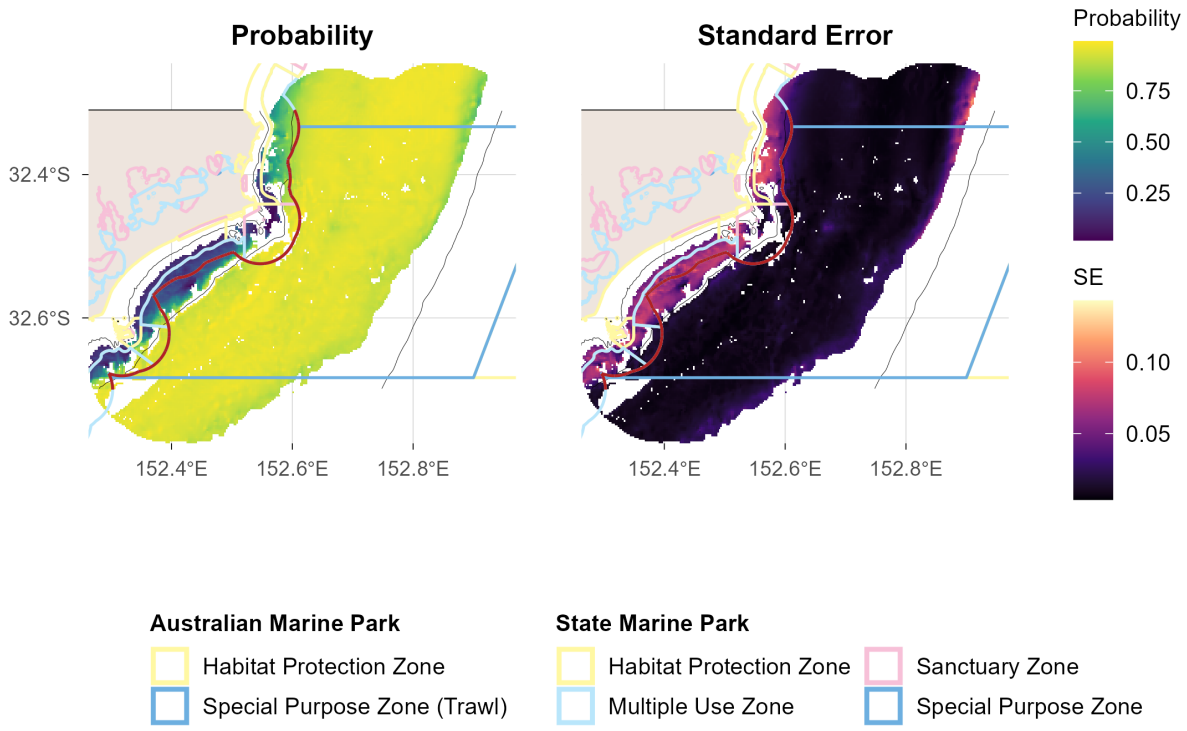


Figure A.4.2: Sediment: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30, 70 and 200 m are shown.

Sessile invertebrates

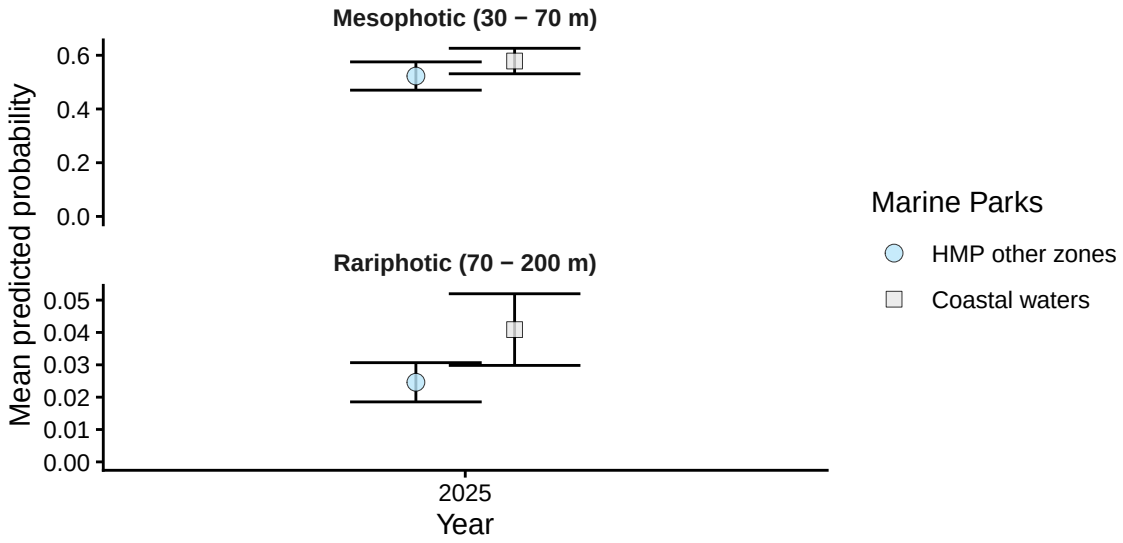


Figure A.5.1: Sessile invertebrates: temporal plots by depth contour from stereo-BRUV groundtruthing. HMP = Hunter Marine Park (Commonwealth).

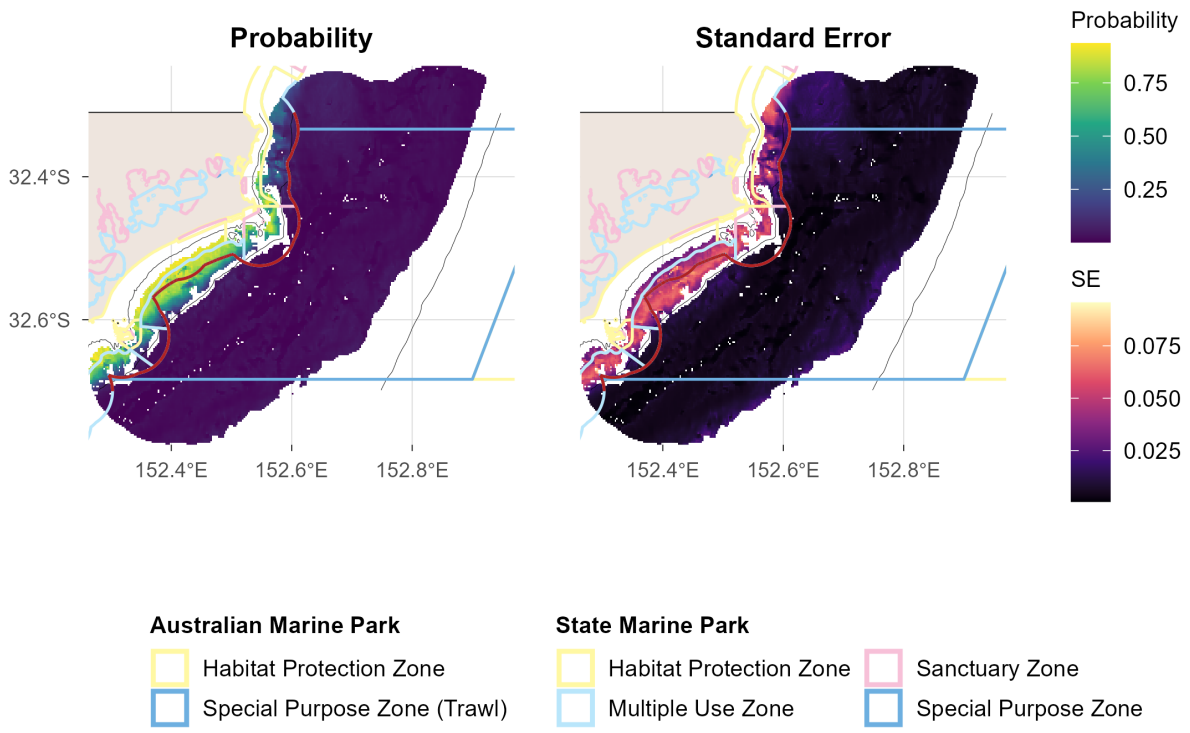


Figure A.5.2: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30, 70 and 200 m are shown.

Benchmarks of demersal fish natural values

Survey effort

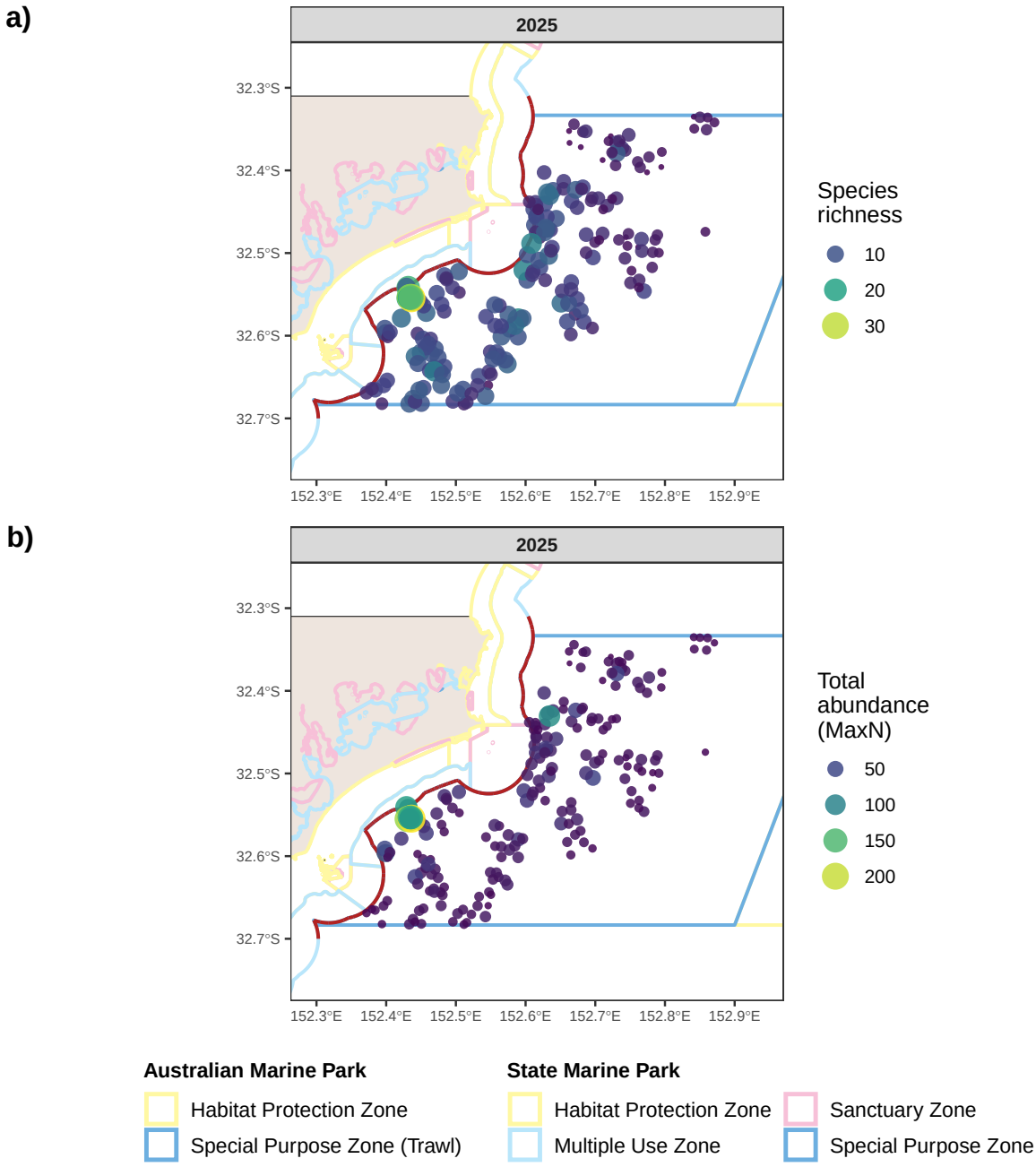


Figure A.6: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

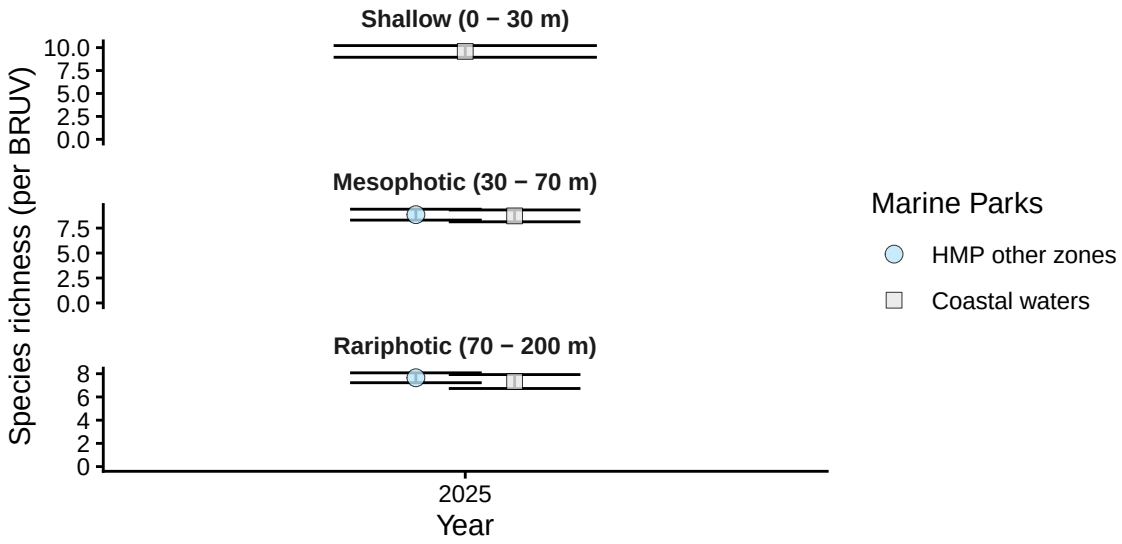


Figure A.7.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. HMP = Hunter Marine Park (Commonwealth).

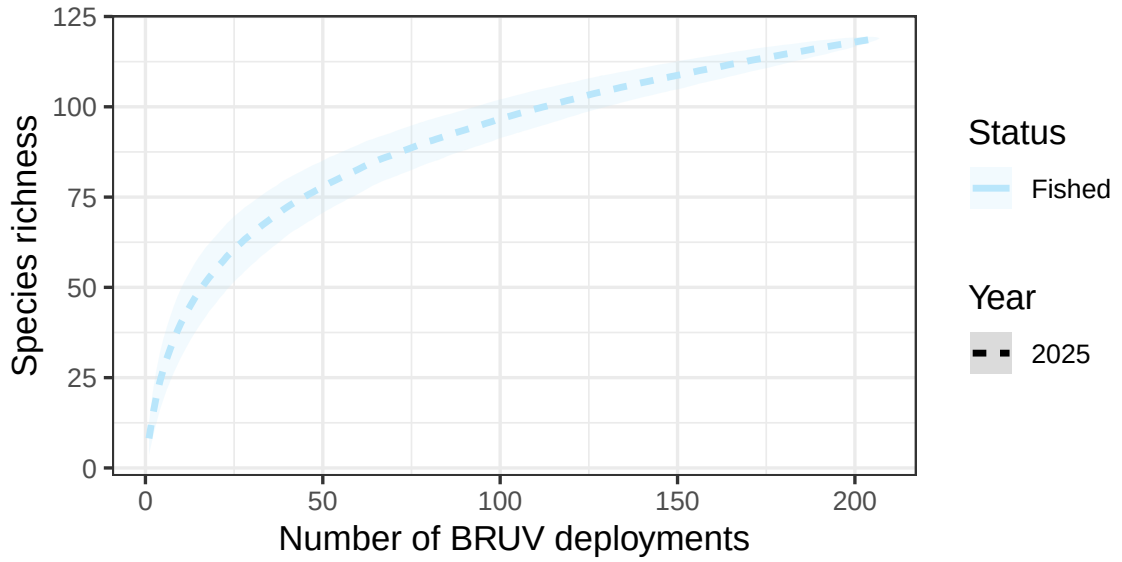


Figure A.7.2: Demersal fish - Species richness: species accumulation curve by year from stereo-BRUV groundtruthing in Hunter Marine Park (Commonwealth).

Species richness

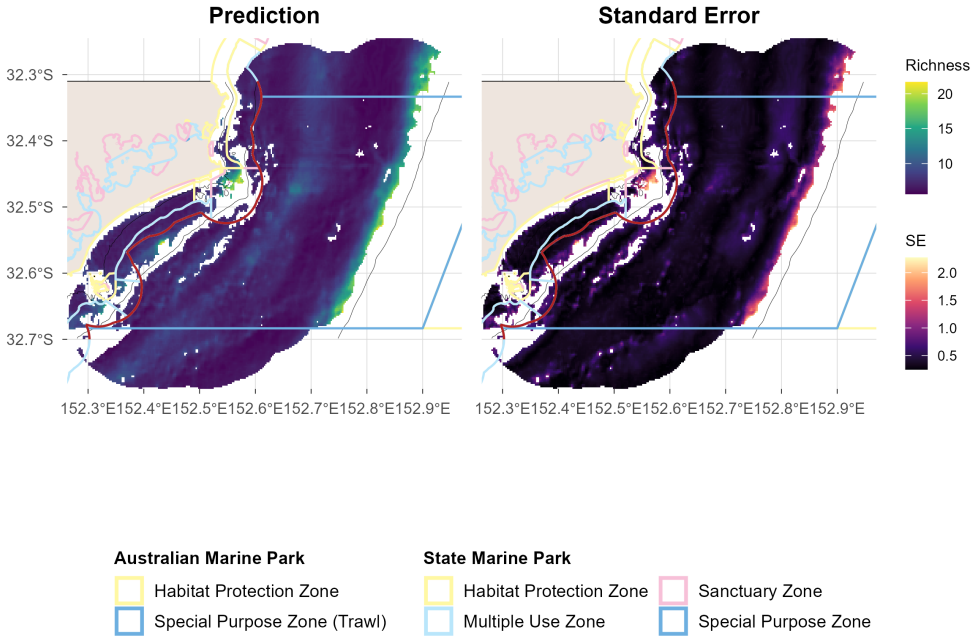


Figure A.7.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30, 70 and 200 m are shown.

Total abundance

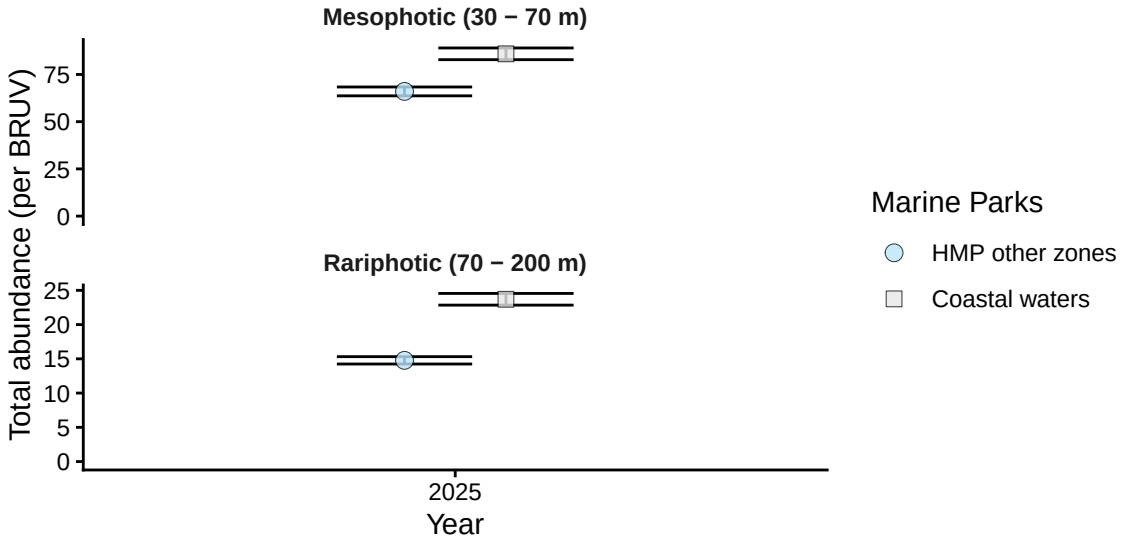


Figure A.8.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. HMP = Hunter Marine Park (Commonwealth).

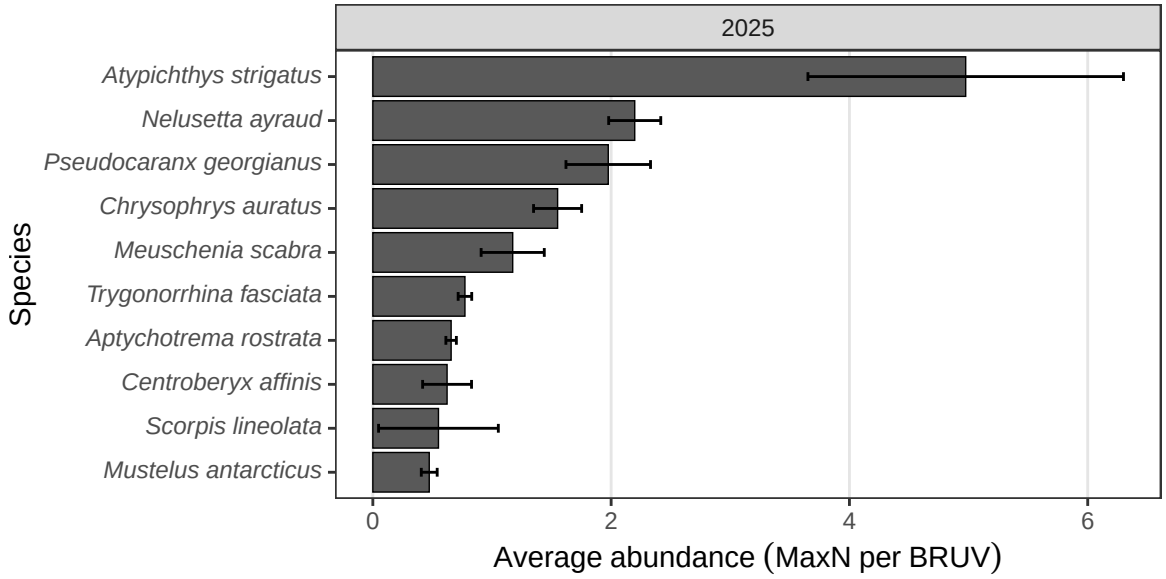


Figure A.8.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing in Hunter Marine Park (Commonwealth).

Total abundance

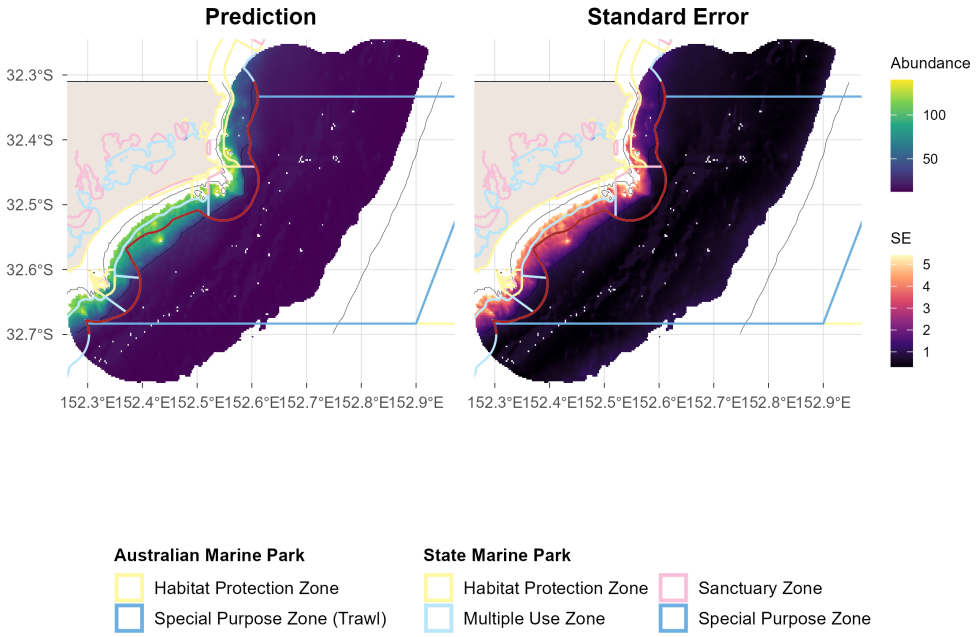


Figure A.8.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30, 70 and 200 m are shown.

Large Reef Fish Index*

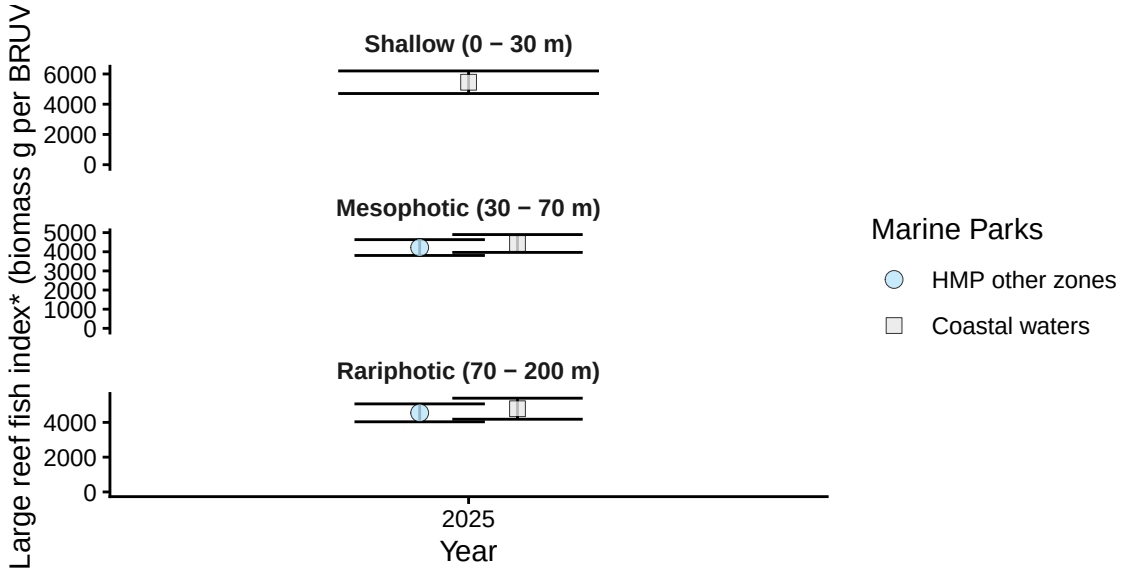


Figure A.9.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. HMP = Hunter Marine Park (Commonwealth), PSGLMP = Port Stephens Great Lakes Marine Park (State). *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

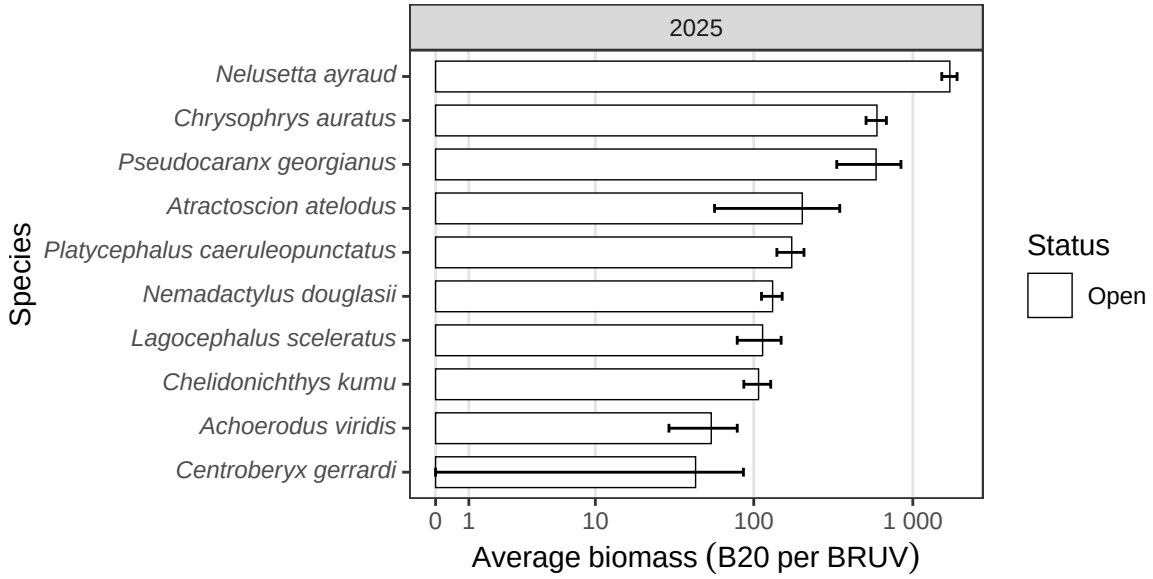


Figure A.9.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing in Hunter Marine Park (Commonwealth).

Large Reef Fish Index*

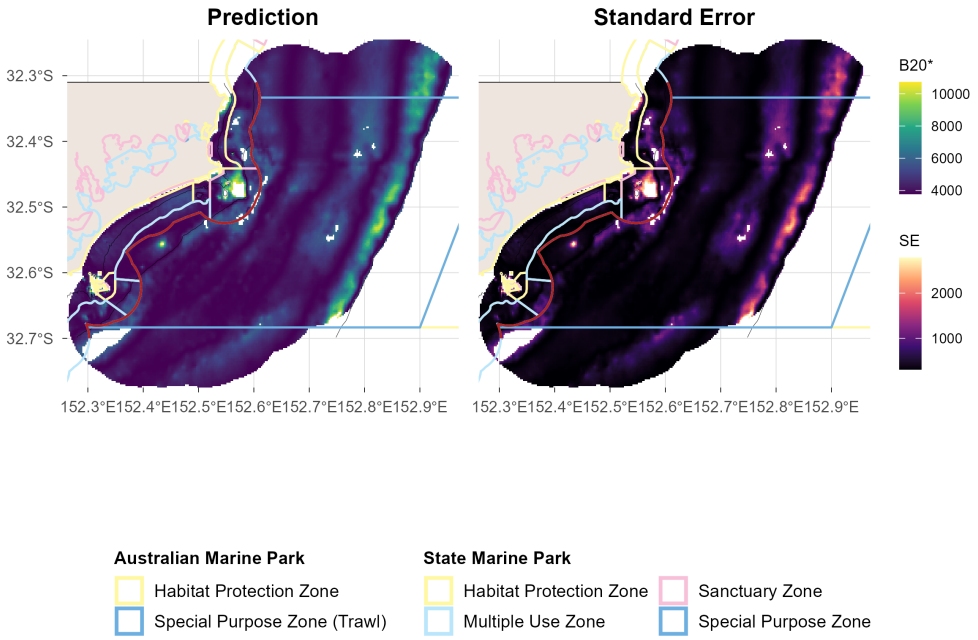


Figure A.9.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30, 70 and 200 m are shown.

Reef fish thermal index

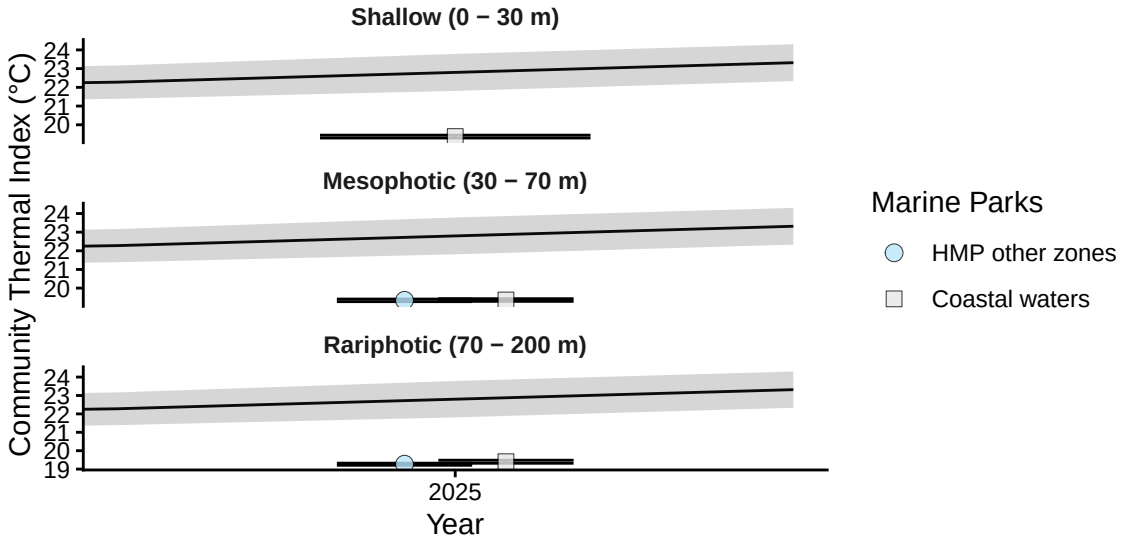


Figure A.10.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing in Hunter Marine Park (Commonwealth).

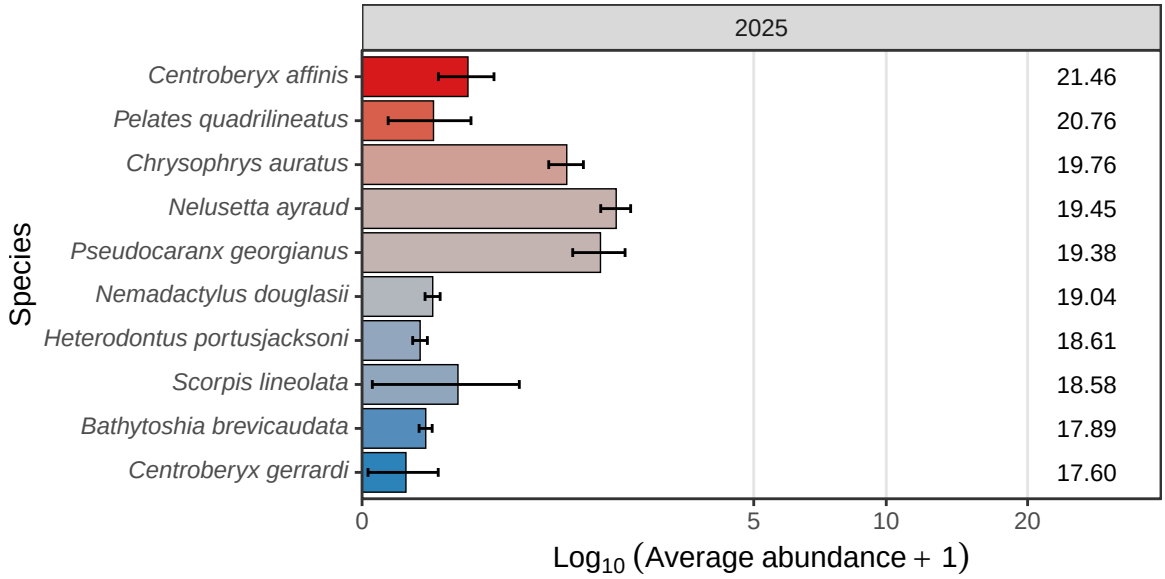


Figure A.10.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing in Hunter Marine Park (Commonwealth).

Reef fish thermal index

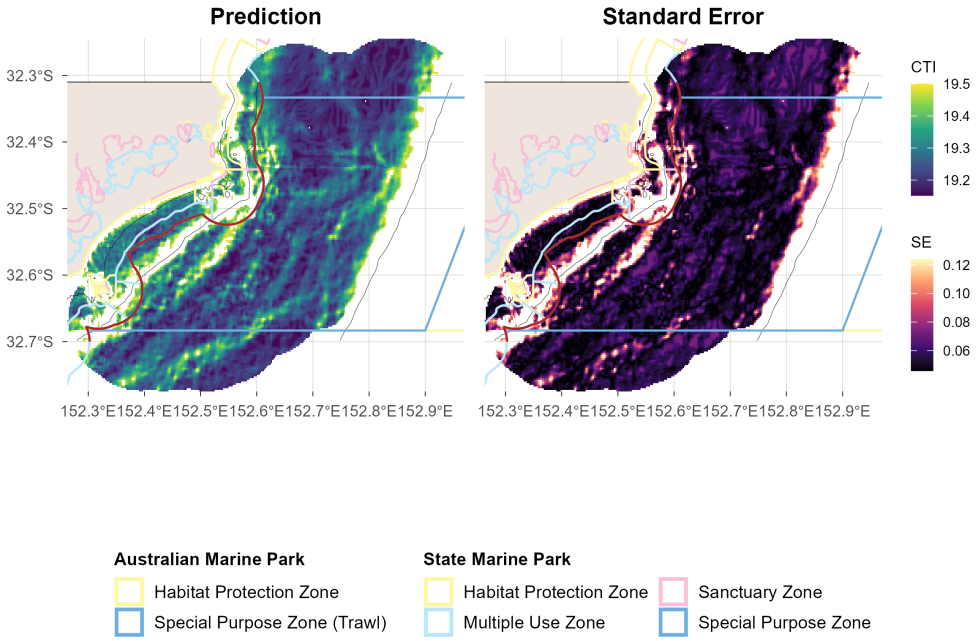


Figure A.10.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contours at 30, 70 and 200 m are shown.

Threatened species

Table A.1: Threatened species recorded in stereo-BRUV imagery across survey, showing IUCN Red List category, the number of samples each species was recorded in, and the total abundance (MaxN) summed across those samples. ^This species is also listed as 'Vulnerable' on the EPBC Act List of Threatened Fauna (NOTE: only east coast population of Greynurse Shark is listed as Vulnerable).

Species	**Common name**	**Family**	**IUCN category**	**Samples**	**Total MaxN**	**Years recorded**
Centroberyx affinis	Redfish	Berycidae	Vulnerable	20	129	2025
Trygonoptera testacea	Common Stingaree	Urolophidae	Near Threatened	16	16	2025
Dentiraja australis	Sydney Skate	Rajidae	Near Threatened	10	10	2025
Achoerodus viridis	Eastern Blue Groper	Labridae	Near Threatened	7	7	2025
Bathytoshia lata	Black Stingray	Dasyatidae	Vulnerable	4	4	2025
Galeocerdo cuvier	Tiger Shark	Carcharhinidae	Near Threatened	4	4	2025
Argyrosomus japonicus	Mulloway	Sciaenidae	Endangered	3	3	2025
Carcharodon carcharias^	White Shark	Lamnidae	Vulnerable	3	3	2025
Squalus montalbani	Philippine Spurdog	Squalidae	Vulnerable	2	3	2025
Carcharhinus limbatus	Common Blacktip Shark	Carcharhinidae	Vulnerable	1	2	2025
Carcharias taurus^	Greynurse Shark	Carchariidae	Critically Endangered	1	1	2025

Threatened shark and ray species



Figure A.11.1: Occurrence of threatened, IUCN Red List category, sharks and rays in stereo-BRUV imagery. Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Other shark and ray species



Figure A.11.2: Occurrence of shark and ray species that are not threatened in stereo-BRUV imagery. Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment, so a panel with no coloured points is a genuine absence rather than an absence of sampling. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Other shark and ray species



Figure A.11.3: Occurrence of shark and ray species that are not threatened in stereo-BRUV imagery (continued). Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment, so a panel with no coloured points is a genuine absence rather than an absence of sampling. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.